

Networks: Costs

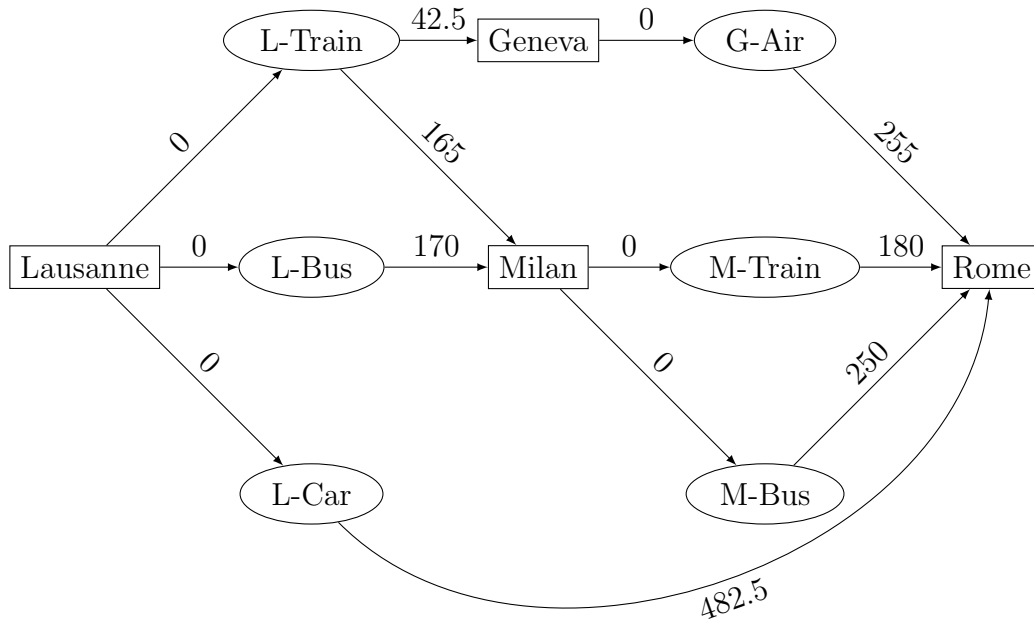
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Solution to practice quiz

First, we construct the network. We define two sets of nodes:

- cities, represented with a rectangle shape in the figure below (Lausanne, Geneva, Milan and Rome),
- transportation modes to leave a city, represented with an elliptic shape in the figure below (Lausanne by train, Lausanne by bus, Lausanne by car, Geneva by air, Milan by train and Milan by bus).

An arc between a city and a transportation mode represents the choice of the transportation mode, and is not associated with any cost. An arc between a transportation mode and a city represents the travel, and is associated with a generalized cost that is calculated below.



- L-car to Rome represents traveling by car between Lausanne and Rome. The cost in CHF is $150 + 9.5 \cdot 30 + 9.5 \cdot 5 = 482.5$.
- L-train to Geneva represents traveling by train from Lausanne to Geneva. The cost in CHF is $20 + 0.75 \cdot 30 = 42.5$.
- L-train to Milan represents traveling by train between Lausanne and Milan. The cost is $60 + 3.5 \cdot 30 = 165$.
- L-bus to Milan represents traveling by bus between Lausanne and Milan. The cost is $20 + 5 \cdot 30 = 170$.
- G-Air to Rome represents traveling by air between Geneva and Rome. The cost in CHF is $210 + 1.5 \cdot 30 = 255$.
- M-train to Rome represents traveling by train from Milan to Rome. The cost is $90 + 3 \cdot 30 = 180$.
- M-bus to Rome represents traveling by bus from Milan to Rome. The cost is $40 + 7 \cdot 30 = 250$.